REPUBLIC POLYTECHNIC

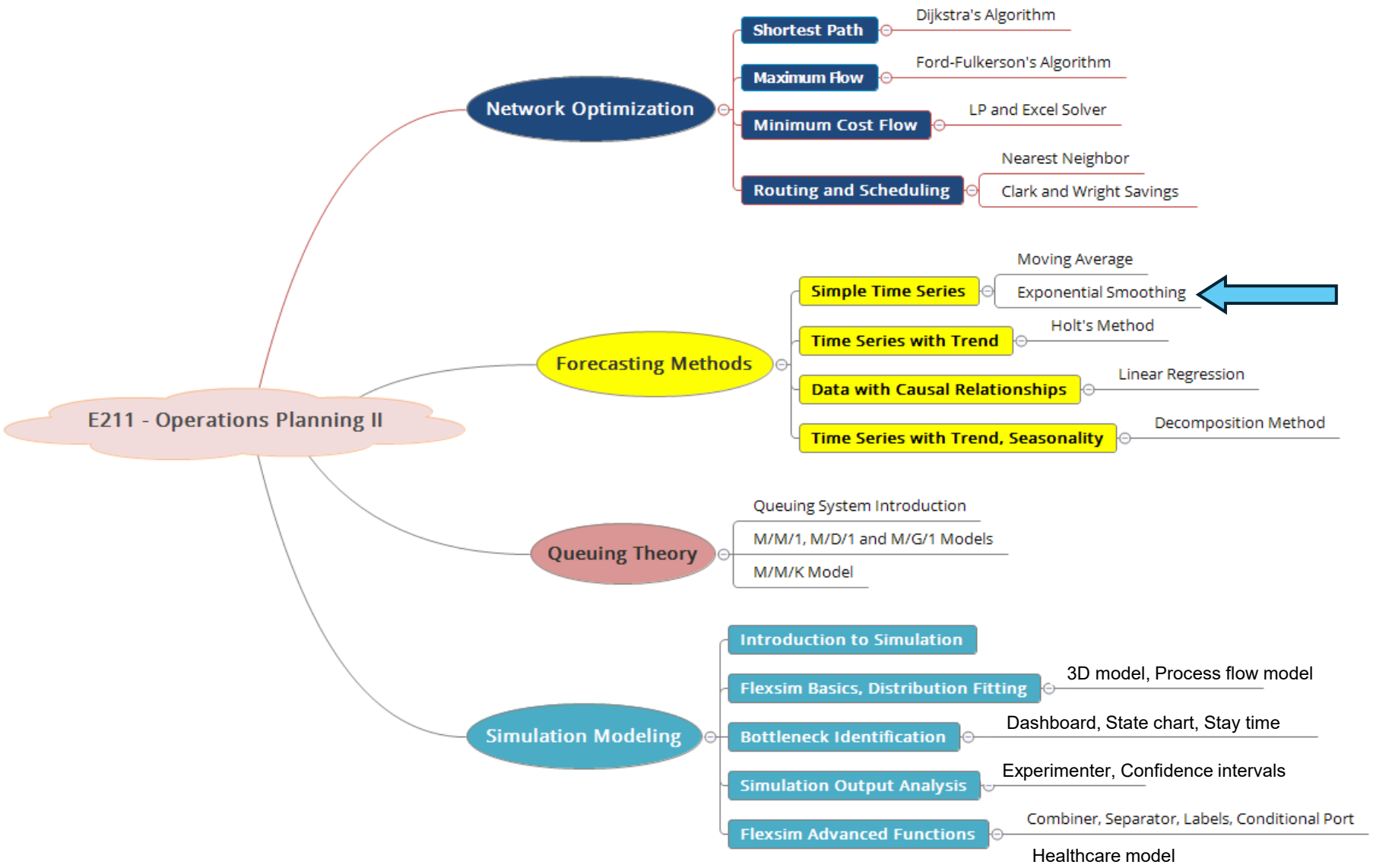
# Lesson 22

# Demand Forecasting: WMA & Exponential Smoothing

**E211 – Operations Planning II**

REPUBLIC POLYTECHNIC

# E211 Topic Tree



# Time Series Forecasting Methods

1. Moving Average (MA)
2. Weighted Moving Average (WMA)
3. Simple Exponential Smoothing (SES)
4. Trend-adjusted exponential smoothing (Holt's method)
5. Decomposition method
    1. Multiplicative decomposition method
    2. Additive decomposition method

# Weighted Moving Average

# Weighted Moving Average (WMA)

- An averaging method that gives more importance (weights) to recent data points
- WMA is more responsive to current trends compared with moving average, which weighs all points equally
- Formula for WMA:

$$F_t = \frac{W_1 A_{t-N} + W_2 A_{t-N+1} + \ldots\ldots + W_N A_{t-1}}{\sum_{i=1}^{N} W_i}$$

Where:

$F_t$ is the forecast in period $t$

$W_i$'s are the individual weights

$N$ is the number of most recent periods used in the forecast.

$A_{t-N}$ = previous $N$ period's actual demand

# WMA Exercise

The following table lists Company ABC's monthly sales of Product A for the past 12 months:

| Month | Sales Figure |
|---|---|
| Oct-20 | 501 |
| Nov-20 | 429 |
| Dec-20 | 352 |
| Jan-21 | 472 |
| Feb-21 | 342 |
| Mar-21 | 367 |
| Apr-21 | 370 |
| May-21 | 437 |
| Jun-21 | 412 |
| Jul-21 | 336 |
| Aug-21 | 388 |
| Sep-21 | 344 |

Apply Weighted Moving Average method with N=3 and weights of 5, 3, 1 to forecast the monthly sales.

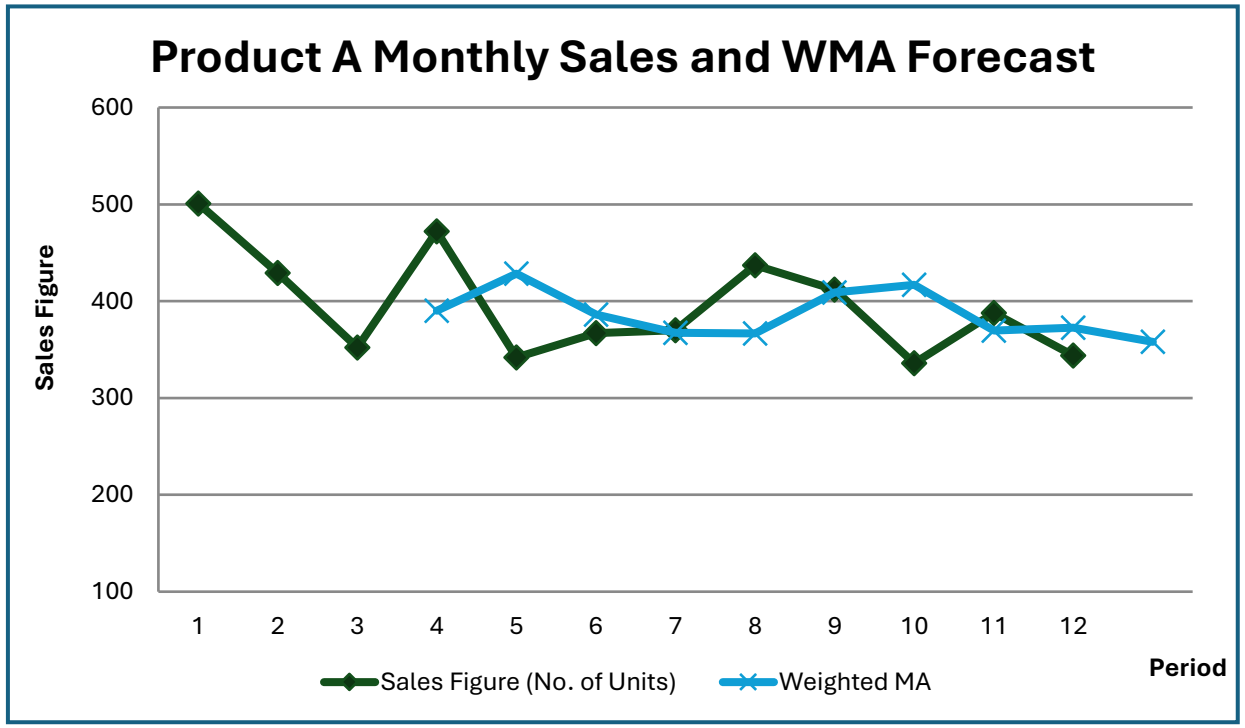
Forecast (Jan-21) = (501*1+429*3+352*5)/ (1+3+5)
= 394.22

1. Work out the forecasts from Feb-21 to Oct-21. Show workings for Oct-21 calculations.
   Oct-21 = [(344*5)+(388*3)+(336*1)]/(5+3+1) = 357.78

2. Calculate the MAD & MAPE for the WMA (N=3) forecasts.

3. Plot the forecast values with the actual sales figures.

# WMA Exercise




1. Forecast for Oct-21 = (344*5 + 388*3 + 336*1)/ (5+3+1) = 357.78

2. MAD = 42.61. MAPE = 11.17%

# Simple Exponential Smoothing

# Simple Exponential Smoothing (SES)

- $F_t = \alpha A_{t-1} + (1 - \alpha) F_{t-1}$

Or

- $F_t = F_{t-1} + \alpha (A_{t-1} - F_{t-1})$

Where:

$F_t$ = forecast for period t

$F_{t-1}$ = forecast for period t-1

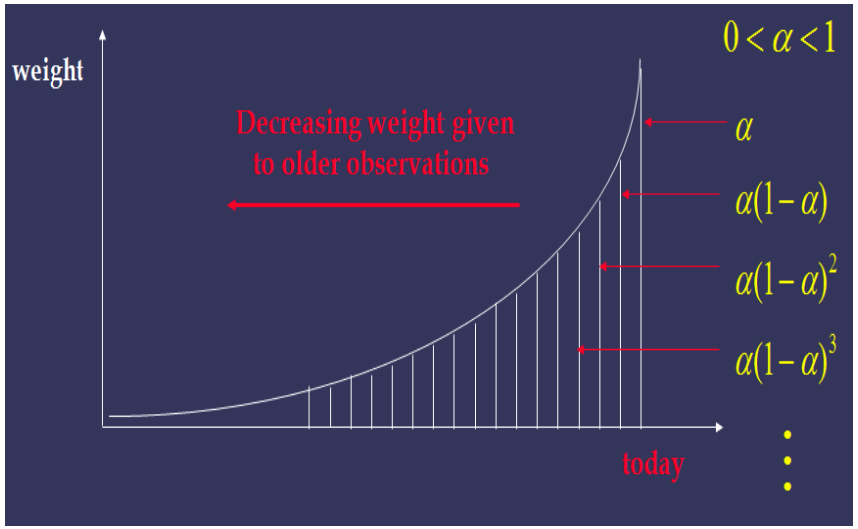
α (alpha) = smoothing constant ($0 \le \alpha \le 1$)

$A_{t-1}$ = actual demand for period t-1

- Expanding the $F_t$ formula,

  $F_t = \alpha A_{t-1} + \alpha(1-\alpha) A_{t-2} + \alpha(1-\alpha)^2 A_{t-3} + \ldots$

- Did you notice that Exponential Smoothing method is a special form of weighted Moving Average Method?



# Simple Exponential Smoothing (SES)

- A form of weighted moving average, widely used method for short-term forecasting
- Requires little record keeping of past data and is best suited for a time series without upward/ downward trend
- Choice of smoothing constant α depends on whether underlying average is likely to change
  - If α value is **small**, response to change is slow, with resultant smooth estimators.
  - If α value is **large**, response to change is fast, with resultant large variability in the output.

➢ Will **lag by 1/α periods if trend is present**

# Simple Exponential Smoothing (SES)

- The current period forecast is always calculated based on previous forecast value → $F_t = \alpha A_{t-1} + (1 - \alpha) F_{t-1}$

How should the initial forecast value be chosen?

- Method 1: First Actual Value

Set the first forecast value $F_1$ = first actual value

- Method 2: Average value of the 1st few actual values

Set the first forecast value $F_1$ = average of 1st N actual values

i.e. $F_1 = \frac{\sum_{i=1}^{N} X_i}{N}$

# SES Exercise

$$F_t = \alpha A_{t-1} + (1 - \alpha) F_{t-1}$$

| Month | Sales Figure |
|---|---|
| Oct-20 | 501 |
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1. Apply Simple Exponential Smoothing to forecast the monthly sales with $\alpha = 0.2$.

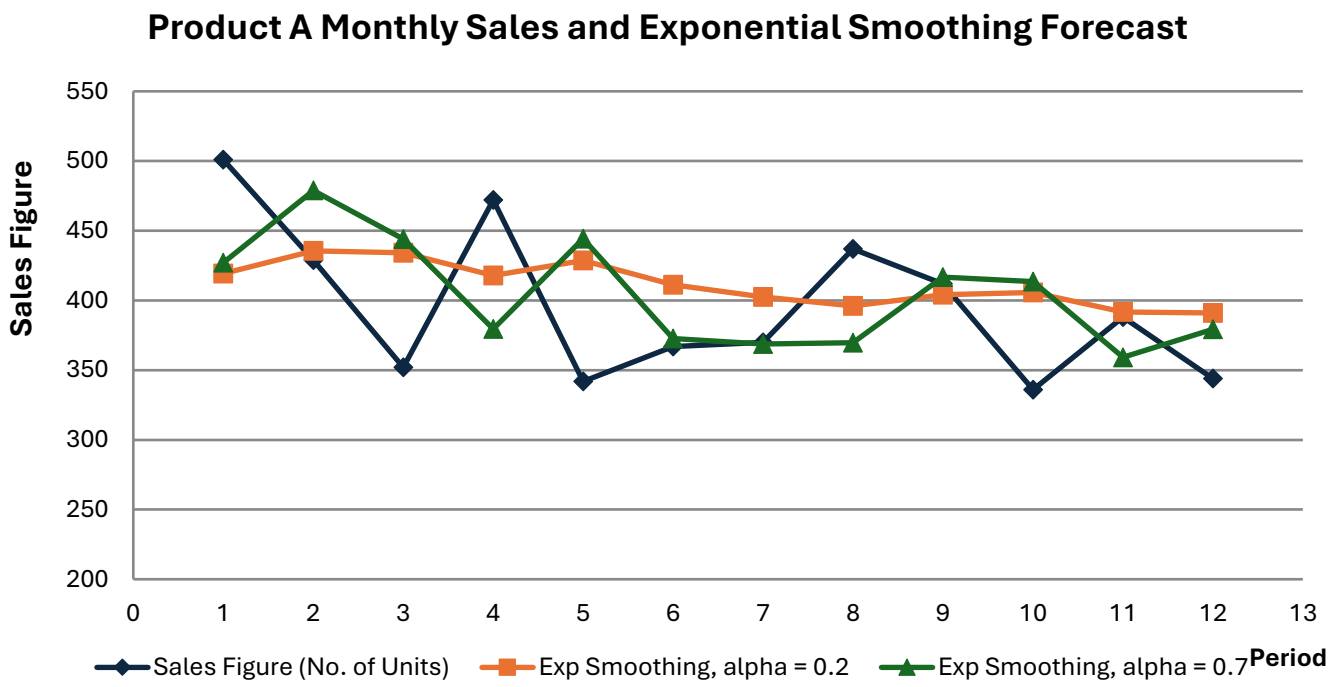
Forecast (Oct-20) = **Average (Oct-20:Feb-21)** = 419.20
Forecast (Nov-20) = 0.2*501 + (1-0.2)*419.20 = 435.56

2. Apply Simple Exponential Smoothing to forecast the monthly sales with $\alpha = 0.7$.

3. Plot the forecasted values with the actual sales figures and comment on the graphs.

4. Calculate MAD and MAPE. Comment on the accuracy of the forecasting methods that you have applied.

# Simple Exponential Smoothing Exercise




- Forecast for Oct-21 is 381.62 (α =0.2) and 354.61 (α =0.7) respectively.
- α =0.7 SES is more responsive to changes in actual sales compared with α =0.2 SES. As α decreases, the forecast becomes less responsive to changes in actual data and the graph becomes smoother.
- α =0.2 SES has an average smaller forecast error compared with α =0.7 SES.

# Exercise: Product B Monthly Sales

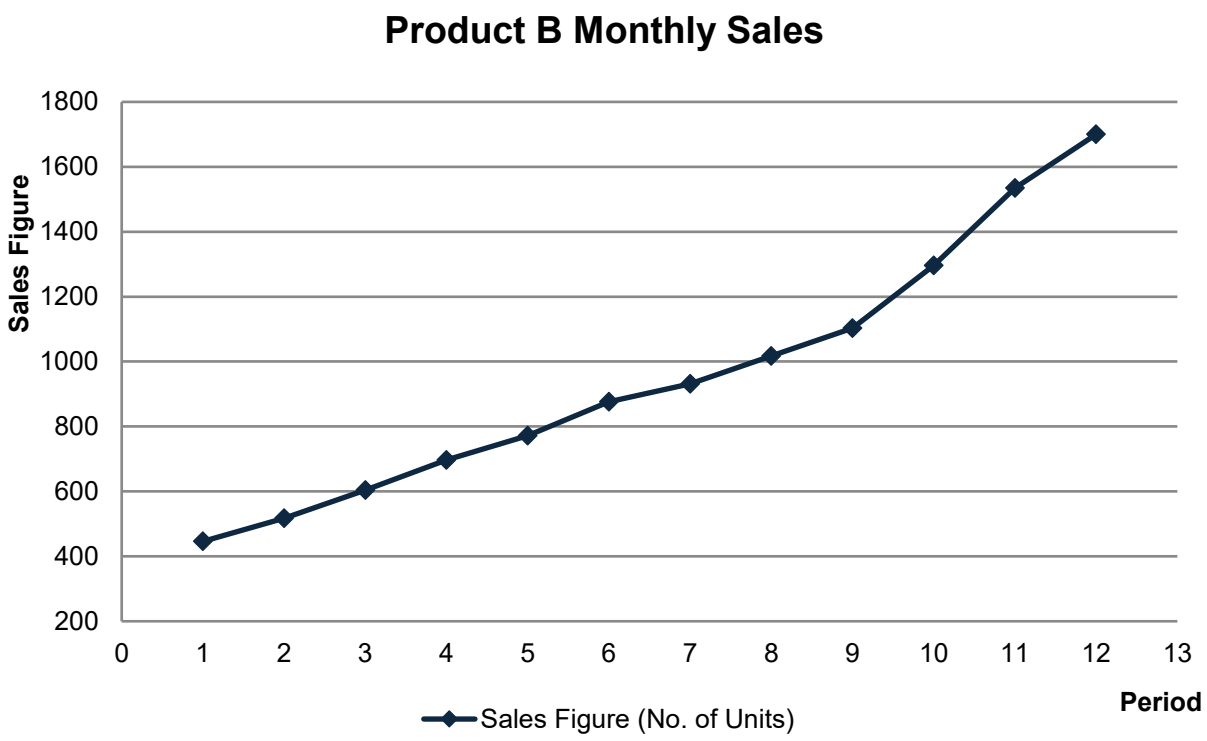
The following table shows Company ABC's monthly sales of Product B for 12 months.

| Month | Sales Figure |
|---|---|
| Oct-20 | 446 |
| Nov-20 | 517 |
| Dec-20 | 604 |
| Jan-21 | 697 |
| Feb-21 | 772 |
| Mar-21 | 876 |
| Apr-21 | 931 |
| May-21 | 1017 |
| Jun-21 | 1103 |
| Jul-21 | 1296 |
| Aug-21 | 1535 |
| Sept-21 | 1700 |

Product B monthly sales data exhibits an upward trend. Will moving average, weighted moving average and simple exponential smoothing methods be suitable for forecasting Product B monthly sales? Investigate.

# Exercise: Product B Monthly Sales

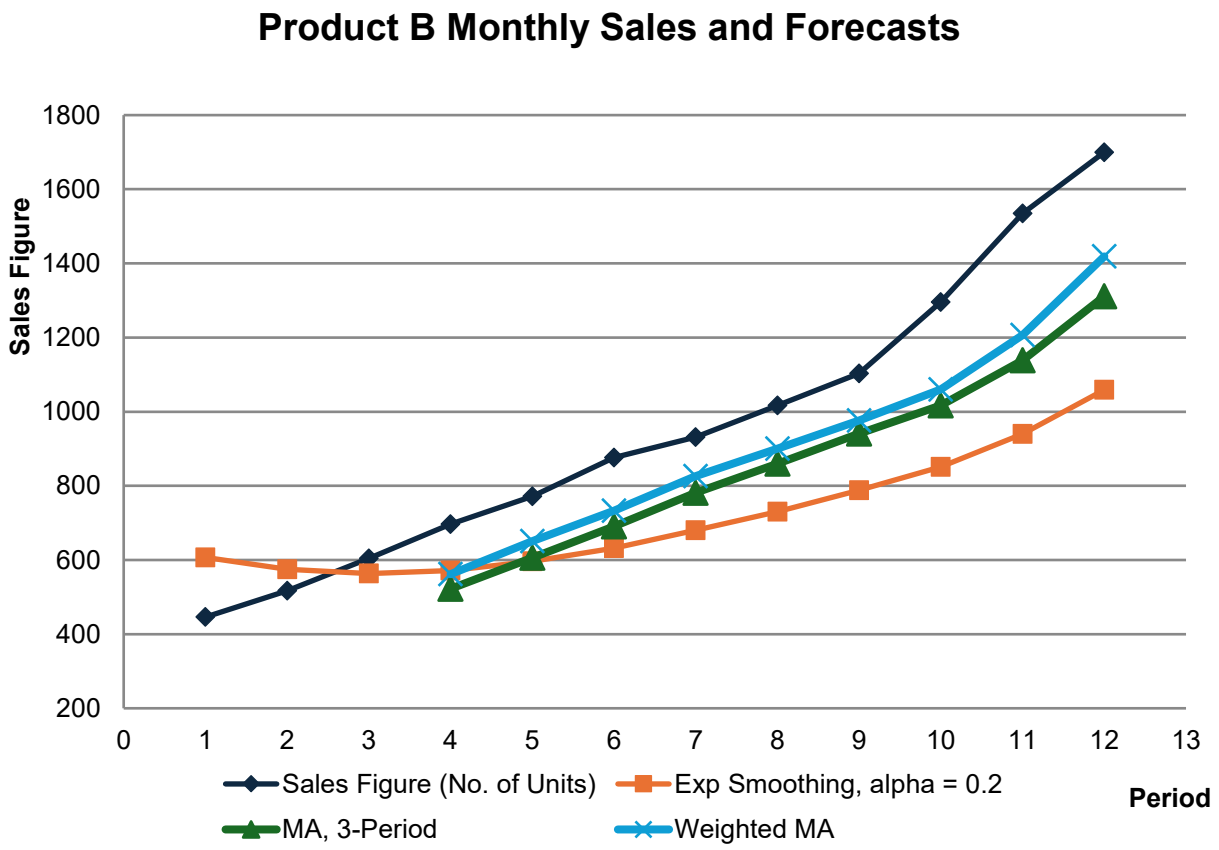
The following graph shows Company ABC's monthly sales of Product B for 12 months.




- Product B Monthly Sales data shows an obvious upward trend.
- Moving average, weighted moving average and exponential smoothing methods are **not suitable** for forecasting Product B sales.
- Forecast from Moving Average method will **lag by (*N*+1)/2 periods**.
  - 3-Period MA will lag by (3+1)/2 = 2 periods
- Simple Exponential Smoothing method will **lag by 1/α periods**.
  - When α = 0.2, the forecasts will lag by 1/0.2 = 5 periods

# Exercise: Product B Monthly Sales

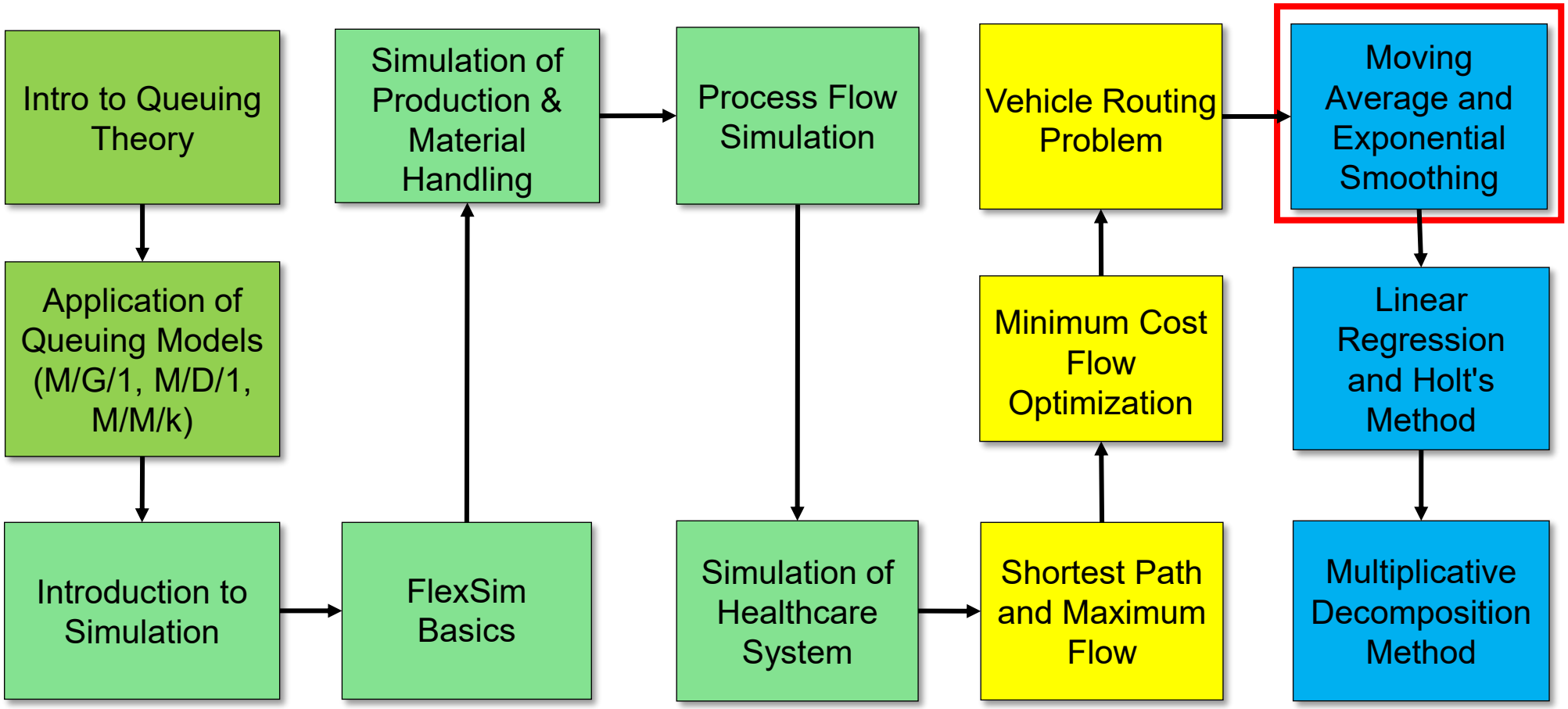
The following graph shows Company ABC's monthly sales of Product B for 12 months.




- The MAPE of Moving Average (N=3) forecast = 20.45%
- The MAPE of 3-Period Weighted Moving Average (Weights are 8, 5, 1) forecast = 15.75%
- The MAPE of Simple Exponential Smoothing (α =0.2) forecast = 26.43%

# Learning Outcomes

At the end of the lesson, students should be able to:

- Apply Weighted Moving Average to forecast a time series.
- Apply Simple Exponential Smoothing to forecast a time series.
- Apply mean absolute deviation and mean absolute percent deviation to determine forecast errors.

# E211 Progression